GGG-B-461a
August 17, 1966
SUPERSEDING
Fed. Spec. GGG-B-461
August 29, 1955

FEDERAL SPECIFICATION

BLOCKS, STEP, MACHINISTS'

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers step blocks used for clamp supports and set-ups on laboratory and machine shop equipment.

1.1.1 *Federal specification coverage.* This specification does not include all types of step blocks (machinists') that are commercially available, but is intended to cover only those generally used by the Federal Government.

1.2 **Classification.**

1.2.1 *Types.* Step blocks covered by this specification shall be of the following types, as specified (see 6.2):

I—Single side.
II—Multiside.
III—Adjustable.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5220

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3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of step blocks which are otherwise in accordance with the requirements of this specification.

3.2 Material. Step blocks shall be of iron or steel, or, if specified (see 6.2), aluminum or seasoned hard maple.

3.2.1 *Design.* Step blocks shall be similar to figures 1 through 3 and as hereafter specified.

3.3 Sizes. Step blocks covered by this specification shall be of the sizes as listed in tables I, II, and III, as specified (see 6.2). If blocks in sizes other than those specified herein are required (see 6.2), they shall otherwise conform to the requirements of this specification.

3.4 Finish. All working surfaces shall be smoothly machined for metal blocks and smooth sanded for wood blocks to a standard commercial finish.

3.5 Identification marking. Step blocks shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.6 Type I, single side. Type I, single side step block shall be right-angle triangular in shape with the horizontal and vertical sides of unequal length. The hypotenuse of the triangle shall form a series of steps. When block is positioned horizontally on either base surface, the step surface in the horizontal plane shall incline 2° ± 1/2° toward the inside corner. A different series of heights shall be available from each face. Standard step blocks shall be similar to figure 1, and shall conform to table I for the size specified (see 6.2).

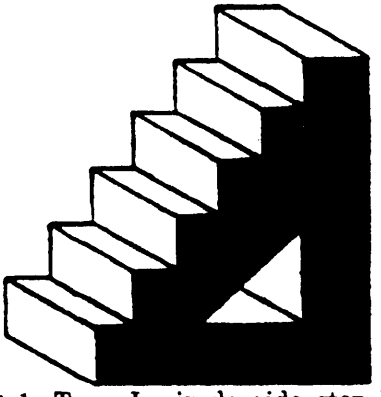


FIGURE 1. Type I, single side step blocks

TABLE I. *Type I, single side step blocks*

Nominal size	Height[1]	Base[1]	Width[1]	Minimum number of steps
Inches	*Inches*	*Inches*	*Inches*	
1-1/2	1-1/2	3/4	1/2	4
2	2	1-1/2	1	4
3	3	2-1/2	1	5
4	4	3	1-1/4	6
6	6	4	1-3/4	6
8	8	6	2	8
9	9	3	2-1/2	4
12	12	4	3	4
18	18	6	4	6
24	24	8	6	8

[1]Tolerances — plus or minus 1/4 inch.

3.7 Type II, multiside. Type II, multiside step blocks shall conform to 3.6 in general body design except that they shall have steps on two, three, or four sides. Each side shall produce a different series of step heights. Steps shall incline 2° ± 1/2° toward the center of the block. Multiside step blocks shall be similar to figure 2 and shall conform to table II for the size specified (see 6.2).

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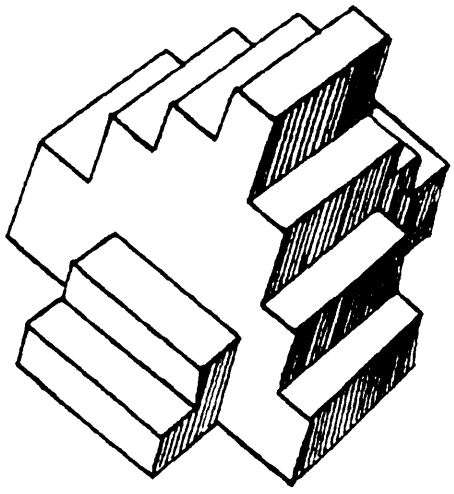


FIGURE 2. Type II, multiside step blocks

TABLE II. *Type II, multiside step blocks*

Nominal size	Base[1]	Height[1]
Inches	*Inches*	*Inches*
3	3	3
6	6	6
9	9	9
12	12	12

[1]Tolerances — plus or minus 1/4 inch.

3.8 Type III, adjustable. Type III adjustable step blocks shall be two identical half blocks, right-angle triangular in shape. The hypotenuse of each triangle shall have a series of teeth or serrations, machined so as to interlock with other machined half blocks. Teeth shall incline 2° ± 1/2° so as to prevent half blocks from slipping apart. Any size half block shall be interchangeable with any other size half block of same manufacturer. When placed together, the two half blocks shall interlock to produce a full block adjustable to varying heights of not more than 1/8-inch increments, and the working surfaces shall be parallel within 1/2°. A different series of heights shall be available from each working face. Adjustable step blocks shall be similar to figure 3, and shall conform to table III for the size specified (see 6.2). Steel for type III blocks shall be casehardened to a minimum depth of 0.005 inch. The surfaces shall be capable of resisting the cutting action of a file.

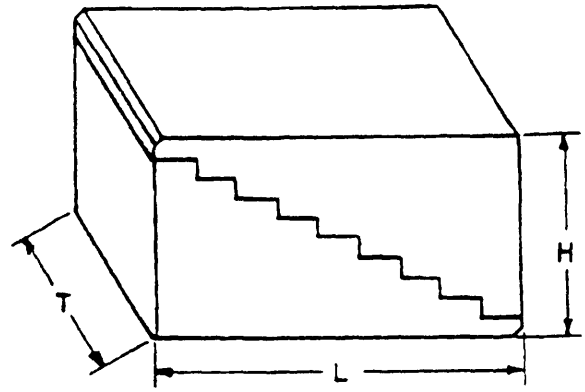


FIGURE 3. Type III, adjustable step blocks

TABLE III. *Type III, adjustable step blocks*

Nominal size	Height[1] H minimum	Length[1] L	Thickness[1] T
Inches	*Inches*	*Inches*	*Inches*
1/2	1/2	1	1/2
1	1	1-1/2	3/4
1-1/2	1-1/2	2	1
2-1/2	2-1/2	3-5/8	2
2-1/2	2-1/2	4-1/8	1
2-3/8	2-3/8	4-1/4	2
3-1/2	3-1/2	4	2
3-5/8	3-5/8	6-3/8	2-1/2

[1]Tolerances—plus or minus 1/4 inch.

3.9 Workmanship. Workmanship shall be first class in every respect. Edges and corners, not necessarily required to be sharp, shall be rounded in conformance with good commercial standards. Step blocks shall have no burrs, excessive roughness, or any other imperfections which may impair their durability and serviceability or create hazardous conditions under normal use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Govern-

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TABLE IV. *Sampling data*

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual and dimensional examination	1 ea.	II	2.5	Percent defective	4.3.1
Testing	1 ea.	II	2.5	Percent defective	4.4.1
Preparation for delivery	One container	S-2	4.0	Defects per hundred units	4.5

ment reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein or, if none, in accordance with this specification.

4.2 **Sampling procedures.** Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table IV.

4.3 **Examination.**

4.3.1 *Visual and dimensional examination.* Each sample unit shall be examined for any nonconformance in design, material, finish, workmanship, and marking.

4.4 **Tests.**

4.4.1 Type III blocks shall be tested for hardness and depth of hardness (see 3.8).

4.5 **Inspection of preparation for delivery requirements.** An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table V. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

TABLE V. *Classification of preparation for delivery defects*

Examine	Defects
Marking (exterior and interior)---	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials-------	Any components missing or damaged.
Workmanship----	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 **Preservation, packaging, and packing.** Unless otherwise specified (see 6.2), the blocks shall be preserved, packaged, and packed in accordance with MIL-H-15424. The levels of preservation and packaging shall be A or C and the level of packing A, B, or C, as specified (see 6.2).

5.1.1 As part of the requirements of section 5 of this specification, table VI is considered as part of, and used in lieu of, that portion of table I of MIL-H-15424 covering machinists' step blocks.

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TABLE VI

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6	Col. 7
Blocks, step, machinists single-side:						
Up to and including 12 in.	P-8, P-9	IIc, IIb	1	----	12	12
13 in. through 24 in.	P-8, P-9	IIb	1	----	6	6
Multiside						
Up to and including 12 in.	P-8, P-9	IIc, IIb	1	----	12	12
Adjustable	P-8, P-9	IIc	1	----	24	24

5.2 Marking.

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The step blocks covered by this specification are intended principally for use in laboratories and by machinists for supporting and setting up job work on various machine shop equipment such as planers, drill presses, and milling machines.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Type and size required (see 1.2.1 and tables I, II, or III).

(*c*) Whether step blocks shall be of aluminum or seasoned hard maple (see 3.2).

(*d*) Sizes other than those specified (see 3.3).

(*e*) Levels of preservation, packaging, and packing required (see 5.1).

MILITARY INTEREST:

Army—MU
Navy—SH
Air Force—67

Review activities:

Army—MU
Navy—SH
Air Force—67

User activity:

Navy—YD

Preparing activity:

Navy—SH

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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